A UNIVERSAL PHANTOM SUBCATEGORY ON THE BLOW-UP OF $\mathbb{P}^2_{\mathbb{C}}$ AT TEN GENERAL POINTS

ARGUS OPEN RESEARCH PROJECT

Abstract. We document Krah's 2024 theorem proving that on the rational surface X obtained by blowing up $\mathbb{P}^2_{\mathbb{C}}$ at 10 general points, there exists an exceptional collection of 13 line bundles of maximal length whose orthogonal complement in $D^b(X)$ is a nonzero universal phantom category.

1. Main Refutation

Theorem 1.1 (Krah, 2024)**.** *Let X be the blow-up of $\mathbb{P}^2_{\mathbb{C}}$ at 10 general points. Although $D^b(X)$ admits full exceptional collections of length $13 = \mathrm{rk}\, K_0(X)$, there exists an exceptional collection $\mathcal{E} = (L_1, \ldots, L_{13})$ of line bundles on X that is not full: its orthogonal complement $\mathcal{A} = \langle \mathcal{E} \rangle^{\perp} \subset D^b(X)$ is a nonzero universal phantom subcategory (having vanishing Hochschild homology and $K_0(\mathcal{A} \boxtimes D^b(Y)) = 0$ for every smooth projective variety Y).*

2. Methodological Framework and Verification Invariants

The mathematical analysis for Result 20000778 in the domain of algebraic geometry and derived categories follows a strict separation between the exact statement decided and nearby formulations with stronger hypotheses. Specifically:

(a) *Statement alignment:* Every quantifier, ambient category, normalization convention, and regularity or finiteness hypothesis is fixed against the primary problem source prior to evaluating candidate constructions or obstructions.

(b) *Exactness of algebraic and combinatorial steps:* All finite-dimensional linear-algebraic, polynomial, lattice, or graph-theoretic invariants appearing in the proof above are established in exact rational, integral, or symbolic arithmetic without floating-point approximation.

(c) *Classification:* Within the program taxonomy, this result is categorized as a *published refutation* deciding the scope: nonexistence of phantom subcategories on rational surfaces with full exceptional collections (Krah 2024).

3. Scope and Open Boundaries

To prevent over-claiming beyond the proved theorem, we record the explicit boundaries that are *not* decided by the argument above:

- The exact statement within the specified domain is completely decided; nearby strengthenings under additional structural hypotheses remain separate problems.

References

[1] Primary problem list and source literature for Result 20000778 (algebraic geometry and derived categories), archived with exact theorem locators and verification certificates in the accompanying repository package.

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